REVISION HISTORY

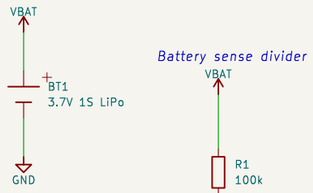
REV	DATE	DESCRIPTION
A	2026-09-07	Initial issue. Point-to-point build, no PCB.

POWER BUDGET (indicative -- measure your own panel)

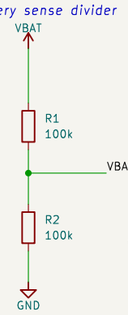
STATE	CURRENT
ESP32-C3 active, radios off	20-25 mA
Backlight at BACKLIGHT_DEFAULT (140/255)	25-50 mA
Backlight at full scale (255/255)	40-80 mA
WiFi TX burst	+100-200 mA pk
Deep sleep after low-voltage cutoff	tens of uA

The backlight dominates. This is why the firmware boots at 140/255 and dims to 40/255 below 3.40 V.

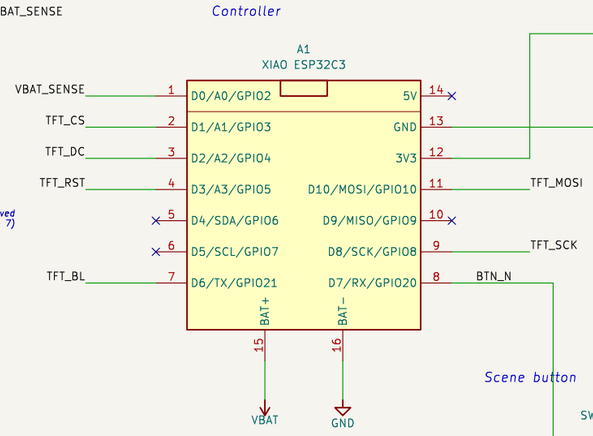
1S LiPo -- charged by A1's on-board charger



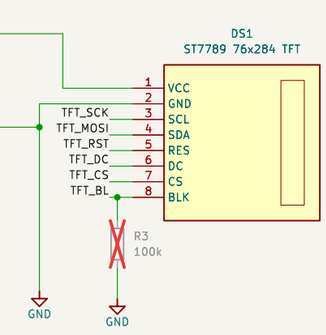
Battery sense divider



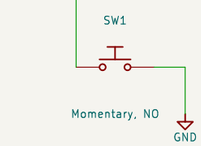
Controller



Display -- 4-wire SPI + PWM backlight



Scene button



WIRE LIST -- point-to-point harness

NET	FROM	TO	NOTES
VBAT	BT1 +	A1 BAT+ (underside pad)	Solder to the pad; no connector
GND	BT1 -	A1 BAT- (underside pad)	
VBAT	A1 BAT+ / BT1 +	R1	Divider top
VBAT_SENSE	R1 / R2 junction	A1 D0 (GPIO2)	Divider tap, ratio 2.0
GND	R2	GND	Divider bottom
+3V3	A1 3V3 (pin 12)	DS1 VCC	Panel logic + backlight LED supply
GND	A1 GND (pin 13)	DS1 GND	Common return; keep it fat
TFT_SCK	A1 D8 (GPIO8)	DS1 SCL	40 MHz -- short, twist with GND
TFT_MOSI	A1 D10 (GPIO10)	DS1 SDA	40 MHz -- short, twist with GND
TFT_CS	A1 D1 (GPIO3)	DS1 CS	
TFT_DC	A1 D2 (GPIO4)	DS1 DC	
TFT_RST	A1 D3 (GPIO5)	DS1 RES	
TFT_BL	A1 D6 (GPIO21)	DS1 BLK	LEDC PWM dimming
BTN_N	A1 D7 (GPIO20)	SW1 pin 1	Active low, internal pull-up
GND	SW1 pin 2	GND	

26-30 AWG stranded silicone hook-up wire. Route the SPI pair and its ground return together and keep them clear of the LiPo. Every joint that flexes when the hat is put on or taken off wants strain relief.

NOTES

1. There is no PCB. This is a point-to-point (flying-lead) build: wire by SIGNAL NAME, not by connector pin number. DS1's pin order varies between vendors -- check it against the silkscreen on your module before cutting wire.
2. Every pin assignment here is duplicated in include/board_config.h and in the -D TFT_* build flags in platformio.ini. All three must agree; CI checks this on every push.
3. A1's on-board charger runs BT1 whenever USB-C is connected. It has NO load disconnect, so the low-voltage cutoff is done in firmware (VBAT_CUTOFF_VOLTS, 3.20 V, then deep sleep).
4. R3 is not fitted by default. Fit it only if the backlight glows faintly during deep sleep: GPIO21 is not an RTC GPIO, so it cannot be held low and goes high-impedance once A1 sleeps.
5. GPIO2 (D0) is a strapping pin that must be high at boot. R1/R2 hold it near 2.1 V with a charged cell, which reads as high, so the divider is safe. If R1/R2 are omitted, set VBAT_SENSE_FITTED = false in board_config.h -- otherwise the floating pin reads as a flat cell.
6. GPIO21 (D6) is also UART0 TX. The ROM bootloader toggles it, so the backlight flickers for about 200 ms at power-on. Cosmetic only.
7. D4/GPIO6 and D5/GPIO7 are left free as an I2C pair for a future sensor (see the roadmap in README.md). D9/GPIO9 is a boot-mode strapping pin and is deliberately left unconnected.
8. SPI runs at 40 MHz, which is only reliable over short leads. Keep SCK and MOSI short, twist them with a ground return, and drop SPI_FREQUENCY to 27 MHz or 20 MHz if the panel shows noise.
9. DS1's MISO is unused and left unconnected (TFT_MISO = -1). The panel is never read back.
10. Assembly: strain-relieve the display flex, and keep BT1 out of the crown of the hat where it can be squashed or punctured.

github.com/andrewaarestad/hat-screen
Pin assignments must match include/board_config.h and platformio.ini.
Point-to-point wiring, no PCB. Wire by signal name; see the notes.

Sheet: /
File: hat-screen.kicad_sch

Title: hat-screen -- wearable ST7789 strip display

Size: A3	Date: 2026-09-07	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1